

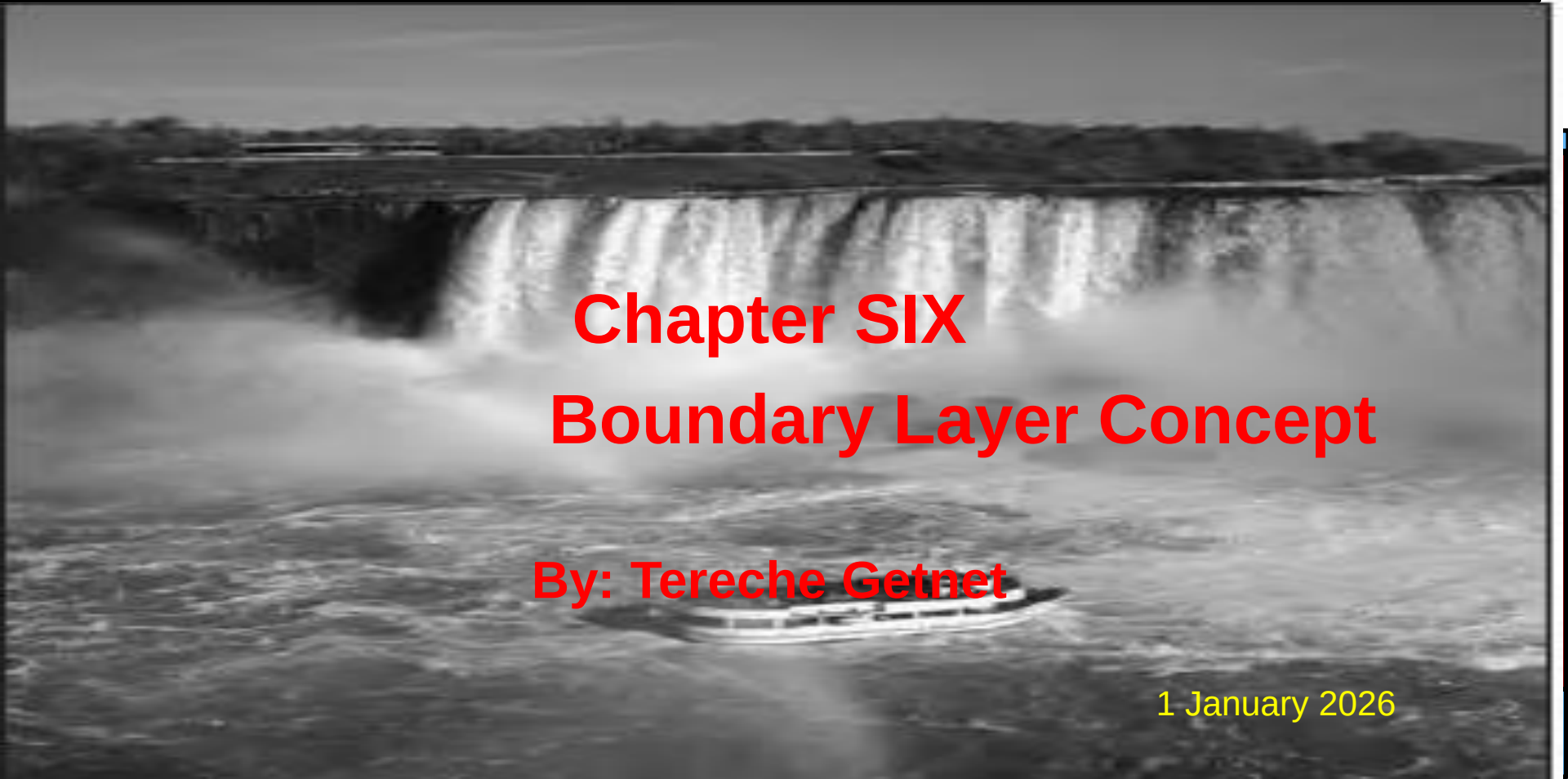


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Faculty of Mechanical and Industrial Engineering

Fluid Mechanics (MEng3133)



Chapter SIX Boundary Layer Concept

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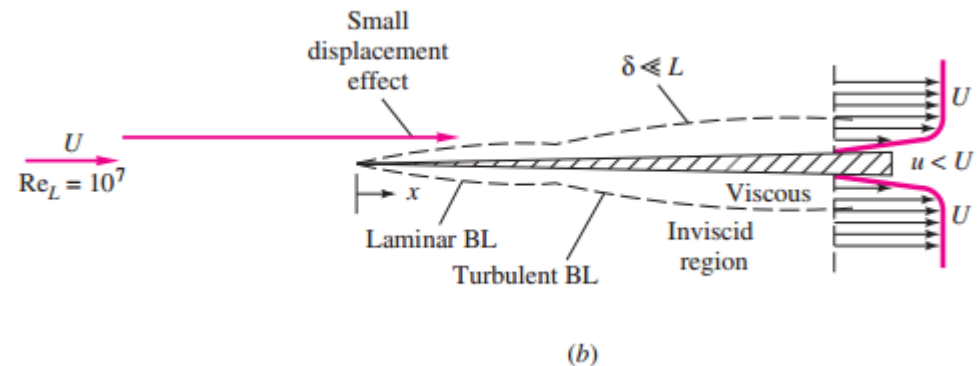
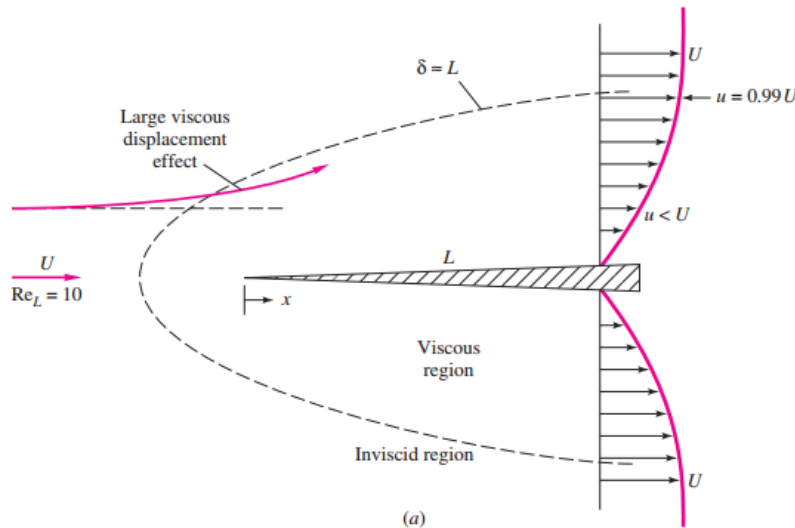
Outline

1. Introduction
2. Reynolds Number and Geometry Concept
3. Momentum Integral Equations
4. Boundary Layer Equations
5. Flow Over A Flat Plate
6. Flow Over Cylinder
7. Pipe Flow
 - Fully Developed Laminar Pipe Flow
 - Turbulent Pipe Flow
 - Losses in Pipe Flow



Reynolds Number and Geometry Effects

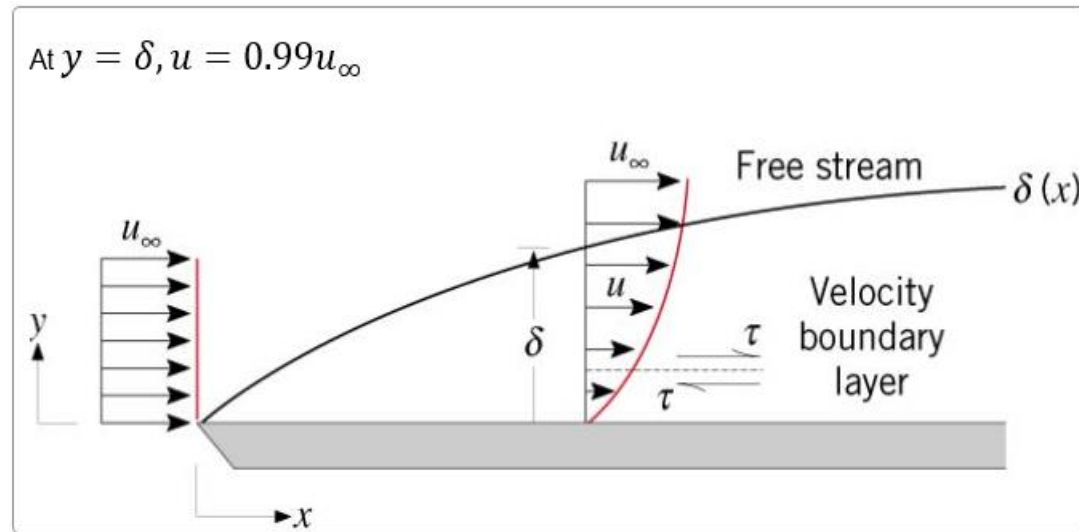
- The technique of boundary layer (BL) analysis can be used to compute viscous effects near solid walls and to “patch” these onto the outer inviscid motion. This patching is more successful as the body Reynolds number becomes larger.





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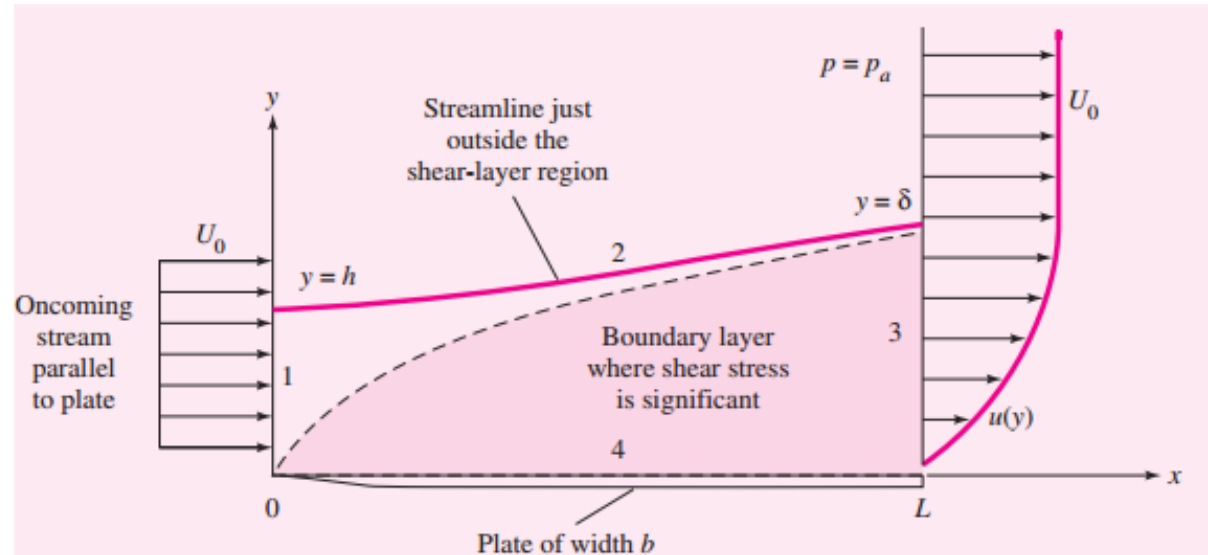
- A high-Reynolds-number flow (Fig. b) is much more amenable to boundary layer patching, as first pointed out by Prandtl in 1904. The viscous layers, either laminar or turbulent, are very thin, thinner even than the drawing shows.
- We define the boundary layer thickness δ as the locus of points where the velocity u parallel to the plate reaches 99 percent of the external velocity U .





Momentum Integral Estimates

The drag force on the plate is given by the following momentum integral across the exit plane: **Ref. from Example 3.11**



$$\begin{aligned} \sum F_x = -D &= \rho \int_1 u(0, y)(\mathbf{V} \cdot \mathbf{n}) dA + \rho \int_3 u(L, y)(\mathbf{V} \cdot \mathbf{n}) dA \\ &= \rho \int_0^h U_0(-U_0)b dy + \rho \int_0^\delta u(L, y)[+u(L, y)]b dy \end{aligned} \quad \rho \int_{CS} (\mathbf{V} \cdot \mathbf{n}) dA = 0 = \rho \int_0^h (-U_0)b dy + \rho \int_0^\delta ub dy \big|_{x=L}$$

$$U_0 h = \int_0^\delta u dy \big|_{x=L} \quad (2)$$

$$D = \rho U_0^2 b h - \rho b \int_0^\delta u^2 dy \big|_{x=L}$$

$$D = \rho b \int_0^\delta u(U_0 - u) dy \big|_{x=L}$$



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- the convenient form of the momentum thickness θ

$$D(x) = \rho b U^2 \theta \quad \theta = \int_0^{\delta} \frac{u}{U} \left(1 - \frac{u}{U} \right) dy \quad (7.3)$$

- Momentum thickness is thus a measure of total plate drag. Kármán then noted that the drag also equals the integrated wall shear stress along the plate:

$$D(x) = b \int_0^x \tau_w(x) dx$$
$$\frac{dD}{dx} = b \tau_w \quad (7.4)$$

- Meanwhile, the derivative of Eq. (7.3), with $U = \text{constant}$, is



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$$\frac{dD}{dx} = \rho b U^2 \frac{d\theta}{dx}$$

- By comparing this with Eq. (7.4) Kármán arrived at what is now called the momentum integral relation for flat-plate boundary layer. It is valid for either laminar or turbulent flat-plate flow.

$$\tau_w = \rho U^2 \frac{d\theta}{dx}$$

(7.5)



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- **Example;** To get a numerical result for **laminar flow**, Kármán assumed that the velocity profiles had an approximately parabolic shape.

$$u(x, y) \approx U \left(\frac{2y}{\delta} - \frac{y^2}{\delta^2} \right) \quad 0 \leq y \leq \delta(x)$$

- which makes it possible to estimate both momentum thickness and wall shear:

$$\theta = \int_0^{\delta} \left(\frac{2y}{\delta} - \frac{y^2}{\delta^2} \right) \left(1 - \frac{2y}{\delta} + \frac{y^2}{\delta^2} \right) dy \approx \frac{2}{15} \delta$$

$$\tau_w = \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} \approx \frac{2\mu U}{\delta} \quad (7.7)$$



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- By substituting (7.7) into (7.5) and rearranging we obtain

$$\delta \, d\delta \approx 15 \frac{\nu}{U} dx \quad (7.8)$$

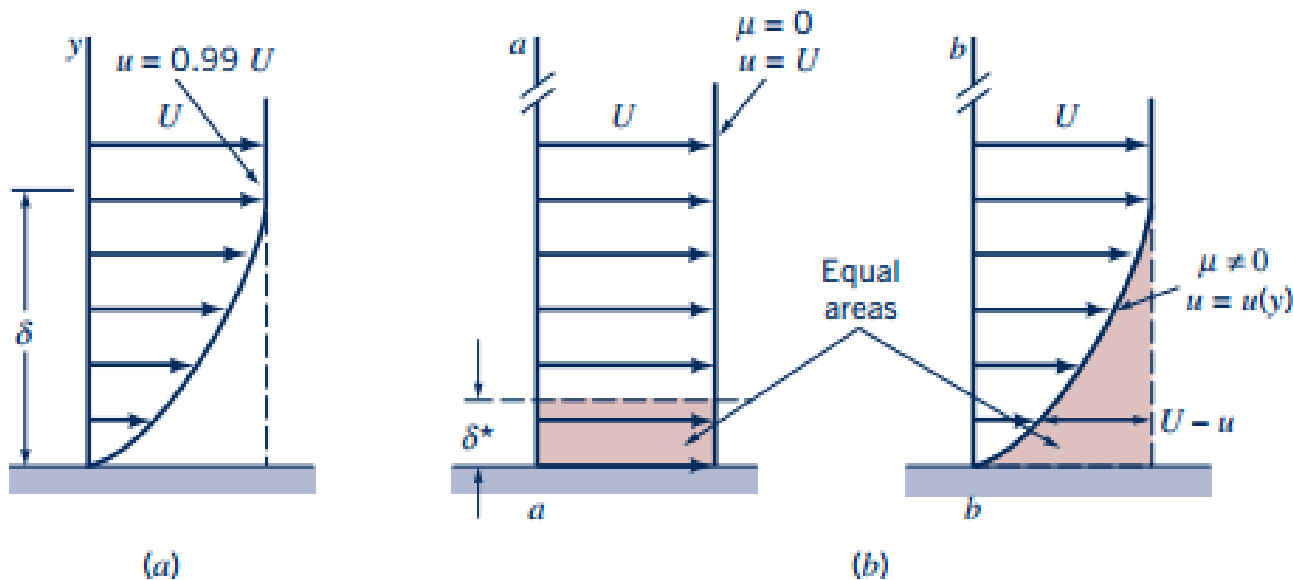
- where $\nu = \mu/\rho$. We can integrate from 0 to x, assuming that $\delta=0$ at $x=0$, the leading edge:

$$\frac{1}{2} \delta^2 = \frac{15\nu x}{U}$$
$$\frac{\delta}{x} \approx 5.5 \left(\frac{\nu}{Ux} \right)^{1/2} = \frac{5.5}{\text{Re}_x^{1/2}} \quad (7.9)$$



- By combining Eqs. (7.9) and (7.7) we also obtain a shear stress estimate along the plate:

$$c_f = \frac{2\tau_w}{\rho U^2} \approx \left(\frac{\frac{8}{15}}{\text{Re}_x} \right)^{1/2} = \frac{0.73}{\text{Re}_x^{1/2}} \quad (7.10)$$





Displacement Thickness

- Another interesting effect of a boundary layer is its small but finite displacement of the outer streamlines. outer streamlines must deflect outward a distance $\delta^*(x)$ to satisfy conservation of mass between the inlet and outlet:

$$\int_0^h \rho U b \, dy = \int_0^\delta \rho u b \, dy \quad \delta = h + \delta^* \quad (7.11)$$

$$\begin{aligned} Uh &= \int_0^\delta (U + u - U) \, dy = U(h + \delta^*) + \int_0^\delta (u - U) \, dy \\ \delta^* &= \int_0^\delta \left(1 - \frac{u}{U}\right) dy \end{aligned} \quad (7.12)$$

Thus the ratio of δ^*/δ varies only with the dimensionless velocity profile shape u/U .



Displacement Thickness

- Introducing our profile approximation (7.6) into (7.12), we obtain by integration this approximate result:

$$u(x, y) \approx U \left(\frac{2y}{\delta} - \frac{y^2}{\delta^2} \right) \quad 0 \leq y \leq \delta(x)$$

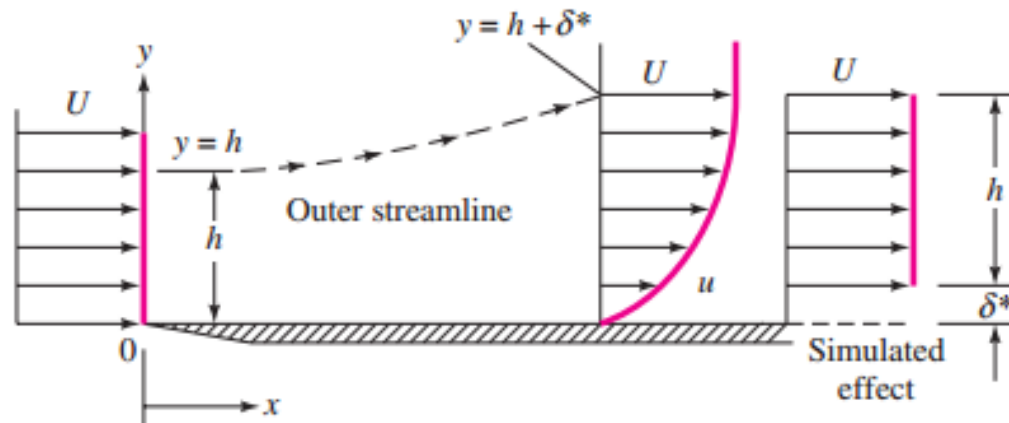
$$\delta^* \approx \frac{1}{3} \delta \quad \frac{\delta^*}{x} \approx \frac{1.83}{\text{Re}_x^{1/2}} \quad (7.13)$$

Since δ^* is much smaller than x for large Re_x and the outer streamline slope V/U is proportional to δ^* , we conclude that the velocity normal to the wall is much smaller than the velocity parallel to the wall. This is a key assumption in boundary layer theory.



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- We also conclude from the success of these simple parabolic estimates that Kármán's momentum integral theory is effective and useful.





Example: The velocity distribution in the boundary layer is given by $\frac{u}{U} = \frac{y}{\delta}$, where u is the velocity y from the plate and $u=U$ at $y=\delta$, δ being boundary layer thickness. Find

- i. The displacement thickness
- ii. The momentum thickness
- iii. The ratio of $\frac{\delta^*}{\theta}$



• solution

Solution:

Velocity distribution: $\frac{u}{U} = \frac{y}{\sigma}$

(i) The displacement thickness δ^*

$$\begin{aligned}
 \delta^* &= \int_0^\delta \left(1 - \frac{u}{U}\right) dy \\
 &= \int_0^\delta \left(1 - \frac{y}{\sigma}\right) dy \\
 &= \left[y - \frac{y^2}{2\sigma} \right]_0^\delta \\
 &= \left(\delta - \frac{\delta^2}{2\sigma} \right) = \delta - \frac{\delta}{2} \\
 &= \frac{\delta}{2}
 \end{aligned}$$

(ii) The momentum thickness

$$\begin{aligned}
 \theta &= \int_0^\delta \frac{u}{U} \left(1 - \frac{u}{U}\right) dy \\
 &= \int_0^\delta \frac{y}{\sigma} \left(1 - \frac{y}{\sigma}\right) dy \\
 &= \int_0^\delta \left(\frac{y}{\sigma} - \frac{y^2}{\sigma^2} \right) dy
 \end{aligned}$$

or

$$\theta = \left[\frac{y^2}{2\sigma} - \frac{y^3}{3\sigma^2} \right]_0^\sigma = \frac{\sigma^2}{2\sigma} - \frac{\sigma^3}{3\sigma^2} = \frac{\sigma}{2} - \frac{\sigma}{3} = \frac{\sigma}{6}$$

(iv) The value of $\frac{\delta^*}{\theta}$

$$\begin{aligned}
 \frac{\delta^*}{\theta} &= \frac{\delta/2}{\delta/6} \\
 &= 3.0
 \end{aligned}$$



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Example 2; The velocity distribution in the boundary layer is given by $\frac{u}{U} = \frac{3}{2} \frac{y}{\sigma} - \frac{1}{2} \frac{y^2}{\sigma^2}$ calculate;

- i. The displacement thickness
- ii. The momentum thickness



- solution**

$$\delta^* = \int_0^\delta \left(1 - \frac{u}{U}\right) dy = \int_0^\delta \left(1 - \frac{3}{2} \frac{y}{\sigma} + \frac{1}{2} \frac{y^2}{\sigma^2}\right) dy$$

$$= \left[y - \frac{3}{2} \times \frac{y^2}{2\sigma} + \frac{1}{2} \times \frac{y^3}{3\sigma^2} \right]_0^\delta$$

$$\left[\sigma - \frac{3}{4} \frac{\sigma^2}{\sigma} + \frac{1}{2} \frac{\sigma^2}{3\sigma^2} \right]$$

$$= \left(\sigma - \frac{3}{4} \sigma + \frac{\sigma}{6} \right)$$

$$\sigma^* = \frac{5}{12} \sigma$$

$$\theta = \int_0^\sigma \frac{u}{U} \left(1 - \frac{u}{U}\right) dy$$

$$= \int_0^\sigma \left(\frac{3}{2} \frac{y}{\sigma} - \frac{1}{2} \frac{y^2}{\sigma^2} \right) \left(1 - \frac{3}{2} \frac{y}{\sigma} + \frac{1}{2} \frac{y^2}{\sigma^2} \right) dy$$

$$= \int_0^\sigma \left(\frac{3}{2} \frac{y}{\sigma} - \frac{9}{4} \frac{y^2}{\sigma^2} + \frac{3}{4} \frac{y^3}{\sigma^3} - \frac{1}{2} \frac{y^2}{\sigma^2} + \frac{3}{4} \frac{y^3}{\sigma^3} - \frac{1}{4} \frac{y^4}{\sigma^4} \right) dy$$

$$= \int_0^\sigma \left[\frac{3}{2} \frac{y}{\sigma} - \left(\frac{9}{4} \frac{y^2}{\sigma^2} + \frac{1}{2} \frac{y^2}{\sigma^2} \right) + \left(\frac{3}{4} \frac{y^3}{\sigma^3} + \frac{3}{4} \frac{y^3}{\sigma^3} \right) - \frac{1}{4} \frac{y^4}{\sigma^4} \right] dy$$

$$= \int_0^\sigma \left(\frac{3}{2} \frac{y}{\sigma} - \frac{11}{4} \frac{y^2}{\sigma^2} + \frac{3}{4} \frac{y^3}{\sigma^3} - \frac{1}{4} \frac{y^4}{\sigma^4} \right) dy$$

$$= \left[\frac{3}{2} \frac{y^2}{2\sigma} - \frac{11}{4} \frac{y^3}{3\sigma^2} + \frac{3}{2} \times \frac{y^4}{4\sigma^3} - \frac{1}{4} \times \frac{y^5}{5\sigma^4} \right]_0^\delta$$

$$= \left[\frac{3}{2} \times \frac{y^2}{2\sigma} \times \frac{11}{4} \times \frac{y^3}{3\sigma^2} + \frac{3}{2} \times \frac{\sigma^4}{4\sigma^3} - \frac{1}{4} \times \frac{\sigma^5}{5\sigma^4} \right]_0^\delta$$

$$\theta = \left(\frac{3}{4} \sigma - \frac{11}{12} \sigma + \frac{3}{8} \sigma - \frac{1}{20} \sigma \right) = \frac{19}{120} \sigma$$



Boundary layer Equations





**Thank you
end of class**